



5.4.5 Cholesky factorization and solving LLS



05.4.5 Solving the Linear Least-Squares problem via Cholesky fa...

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Recall from [Section 4.2](#) that the solution $\hat{x} \in \mathbb{C}^n$ to the linear least-squares (LLS) problem

$$\|b - A\hat{x}\|_2 = \min_{x \in \mathbb{C}^n} \|b - Ax\|_2 \tag{5.4.2}$$

equals the solution to the normal equations

$$\underbrace{A^H A}_B \hat{x} = \underbrace{A^H b}_y.$$

Since $A^H A$ is Hermitian, it would be good to take advantage of that special structure to factor it more cheaply. If $A^H A$ were HPD, then the Cholesky factorization can be employed. Fortunately, from [Homework 5.4.1.1](#) we know that if A has linearly independent columns, then $A^H A$ is HPD. Thus, the steps required to solve the LLS problem [\(5.4.2\)](#) when $A \in \mathbb{C}^{m \times n}$ Are

- Form $B = A^H A$. Cost: approximately mn^2 flops.
- Factor $B = LL^H$ (Cholesky factorization). Cost: approximately $n^3/3$ flops.
- Compute $y = A^H b$. Cost: approximately $2mn$ flops.
- Solve $Lz = y$. Cost: approximately n^2 flops.
- Solve $L^H \hat{x} = z$. Cost: approximately n^2 flops.

for a total of, approximately, $mn^2 + n^3/3$ flops.

Ponder This 5.4.5.1. Consider $A \in \mathbb{C}^{m \times n}$ with linearly independent columns. Recall that A has a QR factorization, $A = QR$ where Q has orthonormal columns and R is an upper triangular matrix with positive diagonal elements. How are the Cholesky factorization of $A^H A$ and the QR factorization of A related?